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CSIS 335
Dr. Teresco
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# Final Project Proposal

The Project Idea:
I will be creating a GPU benchmark program that tests Nvidia's RTX 40 series and RTX 50 series line of graphics cards. RTX 40 series run on Ada Lovelace architecture and RTX 50 series run on Blackwell architecture making the benchmark program I have to write for each slightly different. I will be trying to write in C++ as it seems to be the best choice to have access to GPU control. I will be testing NVIDIA CUDA cores that do large matrix multiplication operations, VRAM capabilities that handles many large copies of data, I will be using the CUDA Toolkit from NVIDIA's dev site that allows me to write, build, and debug my program that will be ran on a NVIDIA GPU. This includes the addition of the nvcc compiler for .cu cuda files.

Milestones:
November: 10 - 25

- Install and understand the NVIDIA Cuda toolkit
- Build program beginning and debug along the way
- Make sure it runs on all 40 series cards first
- Compare advertised card numbers to actual tests ran on card
- Run program many times

November: 25 - December 15

- Make it usable for 50 series cards using the blackwell architecture (might have to be a whole other program)
- Get access to a 50 series card
- Make sure code is understandable and make meaningful comments to explain what each section does
- Add more tests, PCIE transfer, tensor core, etc.
- Make the tests separable if need be so you can run a CUDA test without having to also run a VRAM test via command line options.
- Save results to a csv file for future comparison
- Finish read me to cover what my finished project does

December 15 - 17

- Touch everything up and prepare for submission.